Department of Computer Science and Engineering

Database Management System (DBMS)

Practice Questions

Q1: i) SELECT s.sid, s.sname, c.cname FROM Student s LEFT JOIN Course c ON s.sid = c.cid AND c.cno = (SELECT cno FROM Enrollment WHERE grade = 'A');

Questions: **Convert the following SQL to Relational Algebra.**

ii) SELECT t.tid, t.tname, p.pname, p.salary FROM Teacher t INNER JOIN Payroll p ON t.tid = p.tid AND t.dno = (SELECT dno FROM Department WHERE dname = 'Computer Science');

Questions: **Convert the following SQL to Relational Tuple Calculus.**

iii) A → π sid, sdno, sname, age (Student ⋈ sdno = dno Department)

B → π dname, cdno, clocation, cname (Department ⋈ dno = cdno Course)

Result → π sid, sname, dname, clocation, cname (σ age > 18 (A) ⋈ sdno = cdno B)

Questions: **Convert the following Relational Algebra to SQL.**

Q2: What is a Functional dependency? Find the minimal set of FDs / Minimum cover for F-
F={A → B, AB → C, D → AC, D → E }

Q3: What is physical database tuning? With suitable example explain how Index Sequential Access Method (ISAM) improves the efficiency of retrieval of records from a record-file in DBMS.

Q4: What are serial and non-serial schedules? Consider the transactions T1, T2, T3 and the schedules S1 and S2 given below.
T1 : r1(x); r1(z); w1(z)
T2 : r2(y); r2(z); w2(z)
T3 : r3(y); r3(x); w3(y)
S1 : r1(x); r3(y); r3(x); r2(y); r2(z); w3(y); w3(z); r1(z); w1(x); w1(z)
S2 : r1(x); r3(y); r2(y); r3(x); r1(Z); r2(z); w3(y); w1(x); w2(z); w1(z)
Draw the precidency graph of S1 and S2 and check whether they are **conflict- serializable or not.**

Q5 : Step-by-step optimize the following SQL based using Heuristic Query optimization Technique –

"Select person_name From works, lives, locates_in Where works.person_name = lives.person_name and works.company_name = located_in.company_name and located_in.city = lives.city"

Given the schema of the database: works(personname, companyname, salary); lives(personname, street,city); locates_in(companyname,city)

Q6: What is database normalization? Given a relational schema **R(A, B, C, D, E, F, G,H)** and following functional dependencies –

$A \rightarrow B$, $AC \rightarrow DEFGH$, $F \rightarrow G$, $FC \rightarrow H$, $HCD \rightarrow AEFG$, $FCD \rightarrow AE$

Normalize the above relation. (Assume that the relation is in first NF)

Q7 : Consider the following relations in a schema "Hospital"and answer the queries bellow:

Doctor(<u>SSN</u>, FirstName, LastName, Specialty, YearsOfExperience, PhoneNum)
Patient(<u>SSN</u>, FirstName, LastName, Address, DOB, PrimaryDoctor_SSN)
Medicine(<u>TradeName</u>, UnitPrice, GenericFlag)

Write a query in -

I. Relational algebra to List the trade name of generic medicine with unit price less than Rs50.
II. Tuple calculus List the first and last name of patients whose primary doctor named 'Dr. J. Raghaban'.

SQL to List the first and last name of doctors who are not primary doctors to any patient.

Q8: What is database recovery and why it is essential? Explain the followings with example :

i. Cascading roll-back protocols
ii. Cascadeless rollback protocols

Q9: Why concurrency Control is essential? Explain in details how Time Stamp based concurrency control protocol works.

Q:10 Explain the steps involved in Query processing? With suitable diagram explain different components and their functions of a heuristic Query optimization architecture.

Q:11 Why Query Optimization is essential? What are the job of following components in query optimization architecture model-

Algebric space, method-structure space, cost estimation module